# MYRIAM BENKOUSSA

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## OBJECTIVE

Computer Science master's student with professional experience as a Software Engineer developing and delivering enterprise microservices and AWS cloud solutions in the finance industry. Strong analytical skills and interdisciplinary approach to problem solving, with interest in big data, machine learning, and artificial intelligence, and their industry applications to drive innovation and improve business processes.

## EDUCATION

University of Connecticut, Storrs, CT - Master of Engineering in Computer Science — August 2025-Present
Bryn Mawr College, Bryn Mawr, PA — Class of 2019
Haverford College, Haverford, PA via the Bi-College Consortium - Bachelor's in Computer Science

## PROFESSIONAL EXPERIENCE

**Software Engineer - Credit Card Rewards | JP Morgan Chase** — August 2019-August 2025

- Developed microservices for the Chase Rewards Pay with Points ecosystem hosted in AWS ECS, ensuring seamless integration with payment systems and third-party vendors, and serving millions of credit card customers. Designed scalable, high-performance RESTful APIs to enhance transaction processing and reward redemption efficiency.
- Modernized and migrated legacy Java web applications to AWS EKS and ECS, conducted performance and chaos engineering tests, and implemented Splunk and DataDog for real-time monitoring and observability.
- Designed, developed, and deployed a microservice to automate onboarding for new rewards redemption options, as well as a critical fix for a rewards redemption issue that impacted over 50,000 credit card reward customers.
- Created implementation plans and control procedures for production deployments, provided on-call support, debugged and fixed critical production issues to maintain system stability.
- Served as project manager overseeing a team of contractors to implement data purge scripts in compliance with firmwide data retention policies. Containerized scripts and automated workload execution using Kubernetes and Control-M.
- Collaborated within an Agile team to launch a digital gift card redemption feature for Chase credit card rewards. Designed an event-driven solution of microservices with Kafka and Java-based RESTful APIs hosted in the private cloud.
- Integrated automated vulnerability detection as a part of the CI/CD pipeline of various applications for enhanced security.

**Computer Science Teaching Assistant and Grader | Haverford College** — August 2016-May 2019

- Worked as a TA and grader, and hosted office hours for students for the following courses at Haverford College: Principles of Programming Languages, Principles of Computer Organization, Data Structures, Introduction to Computer Science.

**Software Engineer Intern - Credit Card Rewards | JP Morgan Chase** — June 2018-August 2018

- Worked as a part of an Agile team of interns to develop software solutions to migrate the Chase Loyalty Solutions email service from a monolithic application to a self-contained microservice in the private cloud platform.

**Lead Developer - Digital Scholarship Summer Fellowship | Bryn Mawr College** — June 2017-August 2017

- Worked in a team of fellows to design, develop and implement a data visualization project for large datasets of college archives using Python, JavaScript, D3, and jQuery. Normalized and enhanced datasets using R and OpenRefine.

## PROJECTS AND RESEARCH

- Human activity recognition (HAR) classification using PyTorch, surpassing leading model on HuggingFace
- Computer server anomaly detection application using Python
- Twitter stock data sentiment analysis using Python
- Undergraduate Thesis - completed computer science senior thesis research advised by a Haverford College faculty member.
  Risk Mitigation in Public Key Infrastructure (PKI) Implementation within the Internet of Things (IoT)

## SKILLS

- **Languages:** Python, Java, SQL, Javascript/HTML/CSS
- **Frameworks and Libraries:** PyTorch, TensorFlow, NumPy, Pandas, scikit-learn, Spring/Spring Boot, Spring Batch
- **Tools:** Git, Linux, Docker, Kubernetes, Cloud Foundry, Maven, Jenkins, Postman, Swagger, AWS (EKS, ECS, S3)
- RESTful API proficiency, continuous integration, and continuous delivery/deployment (CI/CD) proficiency
- Agile experience, including roles as both Scrum master and developer on development team, Jira proficiency
- PC proficiency (Microsoft Excel, Word, PowerPoint, Publisher)

## MEMBERSHIPS AND CERTIFICATIONS

AWS Certified Cloud Practitioner - Amazon Web Services — September 2023-Present
Fellow - Rewriting the Code (selective national fellowship for women in technology) — June 2018-Present
Member - Algerian American Association for Science Technology and Entrepreneurship — January 2024-Present
TakeItForward AI/ML Trainer - JP Morgan Chase — July 2024-August 2025
DE&I Council Pillar Lead - JP Morgan Chase — February 2025-August 2025
Software Engineer Program CodeVue Assessor - JP Morgan Chase — September 2021-August 2025